Kasra Ferdowsi

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Education

UC San Diego, *California, USA* 2024

- PhD in Computer Science. Advised by Nadia Polikarpova and Sorin Lerner.

UC San Diego, *California, USA* 2022

- M.S. in Computer Science.

UC Berkeley, *California, USA* 2019

- B.S. in Electrical Engineering and Computer Science. Graduated with Highest Honors.

Professional Experience

Staff Software Engineer *Cubist Inc* 2025 - Present
Senior Software Engineer *Cubist Inc* 2024 - 2025

- Balanced AI-assisted development with careful engineering, using AI agents to improve test coverage, review speed, and code quality while personally owning design and all security- and performance-critical code.
- Contributed over 250 pull requests spanning Rust back-end services, REST and JSON-RPC APIs, AWS CDK infrastructure, TypeScript and Rust SDKs, WebAssembly policies, example programs, and documentation.
- Designed and built an Access Control (ACL) system in Rust providing fine-grained user-friendly control over sensitive resources such as CubeSigner keys, policies, and secrets.
- Co-developed and maintained the Cubist Policy Engine; Implemented a Key-Value store backed by DynamoDB, productionized the Policy SDK, and provided policy design and technical support to customers.
- Led the design, requirements gathering, and implementation for the second version of the *Bascule Drawbridge*, securing the transfer of tokens and assets from malicious actors and bugs.

Research Intern *Microsoft Research* 2023

- Developed an AI harness combining natural language prompts and example values to improve correctness.
- Designed experiments for evaluating the correctness of Python code generated by cutting-edge AI models.

Research Intern *Microsoft Research* 2022

- Designed and implemented ColDeco, the first end-user inspection tool for evaluating code produced by AI models in spreadsheets.
- Conducted a 24-participant controlled experiment to evaluate the interaction.

Software Engineer Intern *Tom Sawyer Software* 2016 - 2019

- Helped design and develop the Tom Sawyer Licensing Server 2.0, a backwards-compatible scalable web server, in Java using a micro-service architecture.
- Co-designed and developed a tool combining Git, Maven and Docker to streamline building and deploying Tom Sawyer Perspectives applications.

Expertise

Programming Languages & Frameworks

- Advanced knowledge of Rust and Python. Fluent in TypeScript, Scala, and Java. Familiar with Haskell, JavaScript, Kotlin, Go, React, and the Spring Framework.

DevOps & Infrastructure

- Advanced knowledge of DynamoDB, Git, Linux, and Bash. Experienced with GitHub Actions, AWS CDK, EC2, Lambdas, Secrets Manager, and other AWS services. Familiar with Docker,

Select Publications

Bottom-up Synthesis of Memory Mutations with Separation Logic
Ferdowsi, K., Peleg, H. ECOOP 2025

- Reformulated bottom-up enumerative synthesis as proof search in Concrete Heap Separation Logic, allowing for efficient and provably correct synthesis in the presence of mutating components.
- Proved the algorithm correct with no purity assumptions, and evaluated an implementation of the algorithm in Scala, showing that it produces high-quality results quickly.

Validating AI-Generated Code with Live Programming
Ferdowsi, K., Huang, R., James, M. B., Polikarpova, N., and Lerner, S. CHI 2024

- Designed and implemented *LEAP*, a novel interaction model for AI programming assistants that support exploring multiple code suggestions through Live Programming.
- Ran a user study with 17 participants, and found that *LEAP* mitigates over- and under-reliance on AI-generated programs, and reduces the cognitive load of validating such code.

The Usability of Advanced Type Systems: Rust as a Case Study
Ferdowsi, K. Research Exam

- Surveyed publications on advanced type systems (Linear types and Region-based memory management) which inspired the Rust programming language, as well as usability studies on Rust itself.
- Drew from broader works in HCI and CS Education to propose paths to addressing some of the difficulties in learning and applying Rust's ownership and lifetime rules.

Investigating the Impact of Using a Live Programming Environment in a CS1 Course
Huang, R., *Ferdowsi, K.*, Selvaraj, A., Soosai Raj, A.G., and Lerner, S. SIGCSE TS 2022

- Implemented a web editor for the Live Programming environment *Projection Boxes*, and developed and deployed a server for collecting usage data from the editor.
- Ran a user study of CS1 students (N > 200), showing the tool's effectiveness for code tracing tasks.

Select Teaching Experience

Teaching Assistant *CSE130: Programming Languages* Winter 2022

- Led discussion sections on Lambda Calculus, Haskell, and implementing an interpreter.

Teaching Assistant *CSE11: Accelerated Intro to Programming* Spring 2020

- Took responsibility for one week of the course in the instructor's absence, leading 2 lectures, and giving an online midterm exam to over 200 students.
- Led the weekly discussion sections, including writing review notes and practice problems tailored to the course, leading live discussions, and creating pre-recorded videos.

Teaching Assistant *CSE131: Compiler Construction* Fall 2019

- Assisted in the design of new course assignments, updated the compilation target of existing assignments from Intel x86 to x86_64, and created tests and autograding infrastructure.